ESE 4125 One-Shot Notes

Safety and Environmental Aspects of Energy Projects · KUET · past papers 2020, 2021, 2022, 2023, 2024 · answers from the course slides

One-shot notes: ESE 4125 Safety and Environmental Aspects of Energy Projects

Core words

Hazard = can hurt (not yet). **Risk** = likelihood × consequence. **Accident** (Heinrich) = unplanned, uncontrolled event giving injury/damage. **Incident** = near miss.

OSHEM = safety (sudden) + health (slow) + environment (surroundings). Safety = unintentional losses; security = intentional.

Hazard categories and sources

Physical, chemical, biological, psychosocial, human-factor ("Please Check Before People Hurt").

Sources: dangerous substances, fire/explosion, machinery, electricity, radiation, overhead work, noise/vibration, temperature/humidity, dust/fumes. Effects: human, economic, environmental.

Control steps: identify → inventory → rank → probability → rank assessment → eliminate/reduce.

Hazard matrix and FDT

Severity 20/15/10/3 × frequency 10/8/6/4/2. 100-200 unacceptable, 60-90 undesirable, 20-50 conditionally acceptable, 6-18 acceptable.

FDT = F T / 2; H = D × FDT. 2022: F 0.5/yr, T 1 week, D 1/yr → FDT 0.0048 ≈ 0.005, H = once in 200 yr.

Thermal and electrical

Heat: cramps (salt) → exhaustion (water) → stroke (thermostat fails, emergency). Cold: hypothermia (core < 35 °C), frostbite (local freezing).

Shock severity: voltage, body resistance, path, duration. 10-20 mA cannot let go; 50-100 mA fibrillation. Protection: insulation, guarding, grounding, fuses/CB/GFCI, lockout, PPE, training.

Radiation limit 5 rem/yr whole body (eyes 15, extremities 50). Time-distance-shielding.

Fire

Triangle: fuel, oxygen, heat (+ chain reaction = tetrahedron). Extinguish: starve (fuel), smother (oxygen), cool (heat), inhibit (chain).

Classes: A solids (water), B liquids (foam), C gases, D metals (special powder), E electrical (CO_2/powder), F/K cooking oil (wet chemical). Water not on B, D, E, F.

RACE: rescue, alarm, contain, extinguish. PASS: pull, aim, squeeze, sweep.

Explosions and BLEVE

Deflagration < speed of sound (~8 bar); detonation > sound (~20 bar); dust explosion; VCE (UVCE open, CVCE confined).

BLEVE: fire heats pressurised liquefied-gas tank → vapour-space wall weakens → rupture → liquid flashes → blast + fireball + fragments. Prevent: water deluge, insulation, relief valves, drainage away, spacing, early evacuation.

High-rise fire

Prevent: design, detectors and alarm, alternate power, no temporary wiring. Fight: sprinklers, hydrants, water reserve, main + diesel standby + jockey pumps, firemen lift, drills. Evacuate: fire-proof stairs with fire doors, never lift, crawl low, assembly point.

Hazard identification

Methods: checklist, what-if, PHA, HAZOP, FMEA, FTA, ETA, relative ranking, audits. Drawings: plot plan (spacing), isometric (piping), BFD, PFD (conditions), P&ID (HAZOP input).

PHA: early; hazard-cause-effect-category (I-IV)-measures. HAZOP: node, intention, parameter + guide word (NO, MORE, LESS, AS WELL AS, PART OF, REVERSE, OTHER THAN) = deviation → causes, consequences, safeguards, actions.

Limitations: completeness, reproducibility, inscrutability, relevance of experience, subjectivity.

Anatomy, FTA, ETA

Normal (optimise; process equipment, control system, procedures) → Abnormal (return to normal; shutdown, protective systems) → Emergency (safe state, minimise losses; containment, emergency response).

FTA deductive from top event (AND/OR); ETA inductive from initiating event (success/failure). Top event of FT = initiating event of ET.

Risk management and bow-tie

Steps: describe system → identify hazards → scenarios → consequence + likelihood → evaluate → reduce → monitor.

Hierarchy: elimination > substitution > engineering > administrative > PPE (last line).

Bow-tie: threats → prevention barriers → TOP EVENT → mitigation barriers → consequences; degradation factors and controls.

PPE

Head (hard hat G 2,200 V, E 20,000 V, C none), eye/face (glasses, goggles, face shield, welding helmet), hearing (plugs, muffs; 90 dBA 8 h), respiratory, hands, feet (steel toe), body, fall harness.

Respirators: APR (dust mask, cartridge, gas mask, PAPR) clean the air; SCBA/SAR supply air (O_2 < 19.5 %, IDLH).

Accidents

Heinrich dominoes: ancestry → fault of person → unsafe act/condition → accident → injury. 88 % acts, 10 % conditions, 2 % unavoidable.

Relationships: man-machine (fitness, knowledge, training, maintenance), man-man (dues, participation), man-environment (noise, heat, light, rules).

Report sheet: identification, person, injury/damage, description, causes, actions, sign-off.

Accident statistics

AFR = injuries × 10^6 / man-hours. ASR = days lost × 1000 / man-hours. FAR = deaths × 10^8 / hours.

2021: 600 × 50 × 40 = 1.2 M h × 0.95 = 1.14 M; AFR = 60 × 10^6/1.14 × 10^6 = **52.63**.

HAZAN = quantify how often and how bad. Safety audit: walk-through, comprehensive, specific, internal/external, compliance.

EPP, TBL, committee

EPP levels: 1 local, 2 whole plant, 3 neighbours + outside help, 4 national. Steps: list disasters, severity, level, plan, resources, procedures, declare/terminate, teams, leader, communication, evacuation; drills.

TBL (Elkington 1997): people, planet, profit.

Committee: chairman (manager), member secretary (safety officer), doctor, personnel officer, foremen, worker reps.

Occupational health

ILO/WHO: highest physical, mental, social well-being; adapt work to man. OSH Act 1970 = law; OSHA = agency (standards, inspections). Service (ILO C161): assess, monitor, examine (pre-employment, periodic), first aid, advise, train, records, rehabilitate.

Diseases

Lung: silica → silicosis, asbestos → asbestosis/mesothelioma, coal → pneumoconiosis, cotton → byssinosis, isocyanates/flour → asthma.

Blood: benzene (aplasia, leukaemia), lead (anaemia), aniline/nitrites (methaemoglobinaemia), arsine (haemolysis).

Urticaria = itchy red welts (latex). Pigmentary = darker/lighter skin (tar, UV, plants). Sore eye: VDU, welding, dust, chlorine. Stress: workload, shifts, bullying, insecurity.

Ergonomics

"Fit the job to the worker" (ergon + nomos). Examples: adjustable chair, screen at eye level, neutral wrists, lifting aids, items within reach, breaks/rotation, good lighting.

Exposure limits

TLV (ACGIH, advice) vs PEL (OSHA, law). TWA 8 h/40 h; STEL 15 min, max 4/day, 60 min apart; Ceiling never. IDLH = escape within 30 min.

TWA = $\Sigma C_i t_i/8$. 2024: 1138/8 = **142.25 ppm** < 200, not exceeded.

Chemicals

Hazardous (health or physical danger) ⊃ toxic (poison). Categories: toxic, corrosive, oxidising, dehydrating, reactive, radioactive, flammable; target organs (hepato-, nephro-, neuro-, haemato-toxins).

Routes: inhalation, ingestion, injection, skin/eye. Local vs systemic; acute (short, high) vs chronic (long, low; benzene → leukaemia).

Dose-response: none < A (threshold) < reversible < B < permanent < C < death (D). Lethal, sub-lethal, acute.

MSDS

16 sections: 1 identification, 2 hazards, 3 composition, 4 first aid, 5 fire, 6 release, 7 handling/storage, 8 exposure/PPE, 9 properties, 10 stability, 11 toxicology, 12 ecology, 13 disposal, 14 transport, 15 regulatory, 16 other.

Supplier prepares and supplies; employer keeps current (≤ 3 yr), accessible, trains; worker follows and knows where it is.

Pollution

Disposal: biodegradable, persistent (slow), non-biodegradable. Existence: quantitative (natural, too much) vs qualitative (man-made).

Air: CO/CO_2, SO_2, NO_x, metals, SPM from fuel burning. Water: industry, sewage, farm runoff, oil spills; control: ETP, STP, IPM.

Local (city air, rivers), regional (acid rain, seas), global (greenhouse, ozone, biodiversity).

Greenhouse, acid rain, smog

Sunlight in → Earth emits IR → GHGs (H_2O, CO_2, CH_4, N_2O, CFCs) trap it → warming.

Acid rain pH < 5.6: SO_2/NO_x → H_2SO_4/HNO_3; wet/dry deposition; fish, forests, buildings. Prevent: scrubbers, low-S fuel, catalytic converters.

Smog: London (coal SO_2 + fog, 1952) vs LA photochemical (NO_x + HC + sun → ozone, PAN).

Sustainable development

Brundtland 1987: meet present needs without compromising future generations. Environmental, economic, socio-political pillars. Needs pollution prevention + clean technology. Challenges: population, urbanisation, intensive agriculture, declining resources, altered cycles, social disintegration.

IEE, EIA, EMP

IEE = quick, pre-feasibility; EIA = detailed, feasibility, quantified. Red: IEE → site clearance, EIA → environmental clearance.

EIA: screening → IEE → scoping/ToR → full EIA → review → decision → monitoring/audit. EMP = mitigation + monitoring plan for construction, operation, decommissioning.

Clearance and DoE

Categories Green, Orange-A, Orange-B, Red. General: application (Form-3, NOC) → site clearance → apply EC → ECC → renew 30 days before expiry (Green 3 yr, others 1 yr).

Red: IEE + ToR → site clearance (30 d) → EIA + ETP approval (60 d) → ECC (60 d).

DoE: 1973 water pollution project → 1977 ordinance/board → 1982 DEPC → 1989 DoE under MoEF → ECA 1995, ECR 1997.

ISO 14001

EMS standard, only certifiable one in ISO 14000. PDCA: policy, aspects + legal, objectives, implement, training, documents, emergency, monitoring/audit, corrective action, management review, continual improvement.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
3R	reduce, reuse, recycle
ACGIH	American Conference of Governmental Industrial Hygienists
AFR	accident frequency rate
AIT	auto-ignition temperature
ANSI	American National Standards Institute
APR	air-purifying respirator
ASR	accident severity rate
BFD	block flow diagram
BLEVE	boiling liquid expanding vapour explosion
CAS	Chemical Abstracts Service (registry number of a chemical)
CB	circuit breaker
CCl_4	carbon tetrachloride
CEO	chief executive officer
CFC	chlorofluorocarbon
CNG	compressed natural gas
CPSC	Consumer Product Safety Commission (USA)
CSR	corporate social responsibility
CVCE	confined vapour cloud explosion
dB	decibel (unit of sound level)
dBA	A-weighted decibel (sound level as the human ear hears it)
DDT	dichloro-diphenyl-trichloroethane (an insecticide)
DEPC	Department of Environmental Pollution Control
DIFE	Department of Inspection for Factories and Establishments (Bangladesh)
DoE	Department of Environment (Bangladesh)
DOT	Department of Transportation (USA)
ECA	Environment Conservation Act (Bangladesh, 1995)
ECC	Environmental Clearance Certificate
ECR	Environment Conservation Rules (Bangladesh, 1997; new rules 2023)
EIA	environmental impact assessment
EMP	environmental management plan
EMS	environmental management system
EPA	Environmental Protection Agency (USA)
EPP	emergency preparedness plan
EPS	earnings per share
ESP	electrostatic precipitator
ETA	event tree analysis

ETP	effluent treatment plant
FAR	fatal accident rate
FDT	fractional dead time
FMEA	failure modes and effects analysis
FTA	fault tree analysis
GFCI	ground-fault circuit interrupter
GHG	greenhouse gas
GHS	Globally Harmonised System of classification and labelling of chemicals
HAZAN	hazard analysis
HAZOP	hazard and operability study
HFC	hydrofluorocarbon
HI	hazard identification
HIRA	hazard identification and risk assessment
HR	human resources
ICC	International Chamber of Commerce
ICRP	International Commission on Radiological Protection
IDLH	immediately dangerous to life or health
IEE	initial environmental examination
ILO	International Labour Organization
IR	infrared
ISO	International Organization for Standardization
LC50	lethal concentration for 50 % of a test population
LD50	lethal dose for 50 % of a test population
LEL	lower explosive limit
LNG	liquefied natural gas
LOTO	lockout/tagout
LPG	liquefied petroleum gas
mA	milliampere = 1/1000 ampere
MIC	methyl isocyanate
MoEF	Ministry of Environment and Forest
MoEFCC	Ministry of Environment, Forest and Climate Change
MSDS	material safety data sheet
mSv	millisievert (radiation dose unit)
NDT	non-destructive testing
NEP	National Environment Policy
NFPA	National Fire Protection Association (USA)
NIOSH	National Institute for Occupational Safety and Health (USA)

NO_x	nitrogen oxides
NOC	no objection certificate
NoC	no objection certificate
NRR	noise reduction rating
NSC	National Safety Council (USA)
OEL	occupational exposure limit
OHS	occupational health and safety
OHSAS	Occupational Health and Safety Assessment Series
OSH	occupational safety and health
OSHA	Occupational Safety and Health Administration (US government agency)
OSHEM	Occupational Safety, Health and Environment Management
P&ID	piping and instrumentation diagram
PAN	peroxyacetyl nitrate
PAPR	powered air-purifying respirator
PASS	Pull, Aim, Squeeze, Sweep (using an extinguisher)
PCB	polychlorinated biphenyl
PDCA	Plan-Do-Check-Act
PEL	permissible exposure limit
PFD	process flow diagram
PHA	preliminary hazard analysis
PM	particulate matter ($PM_{2.5}$ = particles below 2.5 micrometres)
PPE	personal protective equipment
ppm	parts per million
QRA	quantitative risk assessment
RACE	Rescue, Alarm, Contain, Extinguish/Evacuate
REL	recommended exposure limit
rem	roentgen equivalent man (unit of radiation dose; 1 rem = 10 mSv)
RFP	request for proposal
SAR	supplied-air respirator
SCBA	self-contained breathing apparatus
SDS	safety data sheet
SO_x	sulphur oxides
SPM	suspended particulate matter
STEL	short-term exposure limit
TBL	triple bottom line
TDI	toluene diisocyanate

TEDE	total effective dose equivalent
TLV	threshold limit value
ToR	terms of reference
TWA	time-weighted average
UEL	upper explosive limit
UV	ultraviolet
UVCE	unconfined vapour cloud explosion
VCE	vapour cloud explosion
VDU	visual display unit (computer screen)
VOC	volatile organic compound
WHO	World Health Organization